

Detecting Fake Instagram Accounts Using Machine Learning

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1. Summary of the Source

The paper "[Detection of Fake Instagram Accounts via Machine Learning Techniques](#)" by Chelas, Routis, and Roussaki from the National Technical University of Athens, was published in the open-access journal Computers (MDPI) in November 2024.

It addresses the automated detection of fake accounts on Instagram. According to the authors, approximately 10% of Instagram's 2.4 billion accounts (as of 2024) are fake. These accounts are exploited for a range of malicious purposes, including fraud, phishing, blackmail, spam, coordinated influence campaigns, and the commercial trade of followers and likes used to manufacture influencer credibility. Despite Instagram's own countermeasures and user-reporting mechanisms, the prevalence of fake accounts remains high, motivating the need for more effective automated detection methods.

From a cybersecurity perspective, fake accounts function as an infrastructure layer enabling a wide variety of social-engineering attacks. A convincing fake persona can befriend a target, extract organisational information, distribute malicious links, or manipulate public discourse. The scale of the problem - potentially 240 million accounts - makes purely manual review impossible, motivating automated ML-based solutions.

The authors also acknowledge important research challenges surrounding the use of publicly available social-media data: GDPR and privacy constraints, bias and incompleteness in self-reported profile data, rapidly changing user behaviour, and ethical concerns around informed consent when collecting user profiles for research purposes.

The Proposed Solution

The paper proposes a supervised machine-learning pipeline based on publicly available account-level metadata. Seven classifiers are compared: Gaussian Naive Bayes (GNB), Random Forest (RF), Decision Trees (DT), Logistic Regression (LR), Multilayer Perceptron (MLP), K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). A key novelty of the paper is its **four-phase experimental framework**, in which three engineered ratio features are added one at a time - allowing the authors to measure the marginal contribution of each feature to every classifier individually. The authors claim that their approach achieves higher or comparable accuracy relative to prior work despite using a smaller dataset and fewer input features - positioning feature engineering as the primary performance driver.

Datasets Used

Two public datasets are combined:

- [InstaFake Dataset](#) - two JSON files containing confirmed fake-malicious and real Instagram accounts. Features include: profile picture, follower/following counts, post count, description length, account privacy status, and username digit count and length.
- [Instagram Fake Spammer Genuine Accounts \(Kaggle\)](#) - two CSV files (train.csv, test.csv) with partially overlapping features, plus four additional columns: fullname words, nums/length fullname, name == username, and external URL.

The datasets are merged after harmonising column names. The four Kaggle-only columns are discarded as they have no equivalent in the InstaFake JSON. The final combined dataset contains eight base features plus three engineered ratio features added incrementally across the four experimental phases.

2. Critical Evaluation

2.1 Main Claims Made by the Authors

- The approach achieves accuracy higher than or comparable to prior studies that used larger datasets and more features.
- Random Forest is the best algorithm for this task, MLP is the second best, SVM is the worst performer.
- Each of the three engineered ratio features progressively improves classification performance across most algorithms.
- The following/followers ratio is the most impactful single engineered feature, producing the most dramatic improvement (especially in GNB and SVM).
- The followers/posts ratio has low correlation with the target label but still improves MLP performance, justifying its inclusion.

2.2 Are the Claims Supported?

Our reproduction study partially confirms and partially challenges these claims:

- **RF is the best model – confirmed:** In our three-model reproduction, Random Forest achieves the highest scores on all metrics (Accuracy = 0.9488, F1 = 0.9140, AUC-ROC = 0.9836), consistent with the paper's conclusion.
- **Progressive improvement per phase - plausible but unverifiable:** Our notebook adds all three ratio features simultaneously rather than sequentially. This was a deliberate design choice prioritising reproducibility and clarity over replication of the paper's exact protocol. Therefore we cannot reproduce the per-phase ablation that is central to the paper's main contribution.
- **Accuracy higher than prior work - difficult to verify:** The paper reports LR reaching 97% accuracy in Phase 4. Our LR implementation achieves only 91.5%. This gap is likely explained by differences in dataset composition, split strategy, and the fact that the paper evaluates on a different (unreported) test set. Without the paper's exact dataset split and preprocessing code, the comparison is inconclusive.
- **SVM and MLP - reproduced:** All seven classifiers from the paper are implemented and evaluated in our notebook. Our results show SVM achieves AUC-ROC = 0.9614, which differs from the paper's near-random result.

2.3 Evaluation Methodology - Strengths and Weaknesses

Strengths: The four-phase ablation design is genuinely informative - evaluating seven classifiers at four feature configurations produces 28 data points and allows causal attribution of performance changes to specific features. Reporting log-loss alongside AUC-ROC is valuable because it assesses probability calibration, not just ranking ability.

Weaknesses:

- **No cross-validation:** The train/test split strategy is not described in sufficient detail - the paper does not state the split ratio, whether stratification was used, or how many runs were averaged. On a small dataset, a single split yields high-variance estimates.

- **No hyperparameter tuning:** All classifiers appear to use default parameters. Comparing un-tuned classifiers conflates algorithmic quality with configuration - particularly problematic for SVM, which is highly sensitive to the kernel choice and regularisation parameter C .
- **Class imbalance not addressed:** The paper acknowledges that SVM's poor performance is partly due to class imbalance but does not apply any remedy (SMOTE, class weighting, threshold calibration) to any model.
- **Inappropriate correlation measure:** The paper uses Pearson correlation to assess feature relevance, presenting a correlation matrix without justification for this choice. Given the extreme outliers in #followers (up to 15M), Pearson severely underestimates its true relationship with the fake label (-0.04). Our Spearman analysis reveals a correlation of -0.62, making it one of the strongest predictors. This methodological choice led the authors to underestimate a key feature.
- **Ratio feature cap for zero-denominator cases:** When followers = 0, the paper caps the ratio at 2000 (the maximum observed in the rest of the data), reasoning that only fake accounts exhibit this behaviour. This introduces an assumption that is not empirically validated and could artificially inflate separability.
- **No statistical significance testing:** Performance differences between phases and between classifiers are reported as point estimates without confidence intervals, making it impossible to determine whether improvements are statistically significant.

2.4 Possible Weaknesses and Limitations

- **Dataset heterogeneity:** The InstaFake JSON and the Kaggle CSV were collected independently, at different times, possibly using different criteria for labelling accounts as fake. Merging them may introduce distributional artifacts - the model may learn to distinguish the two collection sources rather than genuine vs. fake accounts.
- **Temporal invalidity:** Neither dataset carries collection timestamps. Instagram's algorithm, user behaviour, and bot strategies evolve over time. A model trained on historical data may degrade rapidly in production.
- **Adversarial fragility:** The paper does not discuss robustness to adversarial manipulation. An attacker who knows the features can purchase followers (to reduce the following/followers ratio), add a profile picture, or write a short biography to evade the classifier.
- **Information loss during merging:** The four Kaggle-only features (external URL, name == username, fullname words, nums/length fullname) are discarded. Prior work achieved ~92.5% RF accuracy on the Kaggle dataset alone using these features, suggesting they carry discriminative signal.

2.5 Are the Conclusions Justified?

The paper's conclusion that Random Forest is the best algorithm for this problem is supported by the evidence presented and confirmed by our reproduction. The broader claim - that this approach outperforms prior work despite using smaller data and fewer features - is stated but not rigorously demonstrated. The comparison to prior work is informal: the paper lists other studies' best-reported numbers in a table without controlling for dataset, split, or evaluation protocol differences. A fair comparison would require running all baselines on the same dataset. The claim should therefore be treated as suggestive rather than conclusive.

3. Feature Engineering Analysis

3.1 Features Used

The final feature set after all four phases contains 10 columns:

#	Feature	Type	Phase	Notes
1	hasProfilePicture	Binary	1	1 if a profile picture exists
2	nums/length username	Ratio	1	Digit count / username length
3	descriptionLength	Integer	1	Biography character count
4	isPrivate	Binary	1	1 if account is private
5	numberOfPosts	Integer	1	Total media posts
6	numberOfFollowers	Integer	1	Follower count
7	numberOfFollowing	Integer	1	Following count
8	hasDigitsInUsername	Binary	1	Used in the original paper, replaced in our reproduction by the continuous nums/length username ratio, which preserves more information
9	following/followers ratio	Ratio	2	Following / Followers (capped in paper, Laplace smoothing in our reproduction)
10	following/posts ratio	Ratio	3	Following / Posts (capped in paper, Laplace smoothing in our reproduction)
11	followers/posts ratio	Ratio	4	Followers / Posts (capped in paper, Laplace smoothing in our reproduction)

3.2 Transformations and Normalisation

The paper does not specify whether feature scaling is applied. Our notebook applies StandardScaler to Logistic Regression and KNN inputs, which is mathematically correct: KNN's Euclidean distance is dominated by high-magnitude features (e.g., numberOfFollowers, max ~15 million) without normalisation, and LR's L2 regulariser penalises large-coefficient dimensions inequitably. Random Forest is invariant to monotone transformations and is therefore trained on raw values.

No logarithmic or Box-Cox transformation is applied in either the paper or our notebook, despite the severe right-skew in the count distributions (median followers = 269, max = 15,338,538). A log1p transformation would compress extreme outliers and likely improve LR and KNN performance on these features.

3.3 The Ratio Features - Design and Rationale

The three engineered features are the paper's central methodological contribution. Their design reflects well-established bot-behaviour patterns:

- **following/followers ratio (Phase 2):** Fake accounts typically follow many users while attracting few followers - a follow-back growth strategy common in bot networks. The paper notes a critical issue: when followers = 0, the ratio is undefined. The authors cap it at 2000 (the maximum observed value in the rest of the data), reasoning that only fake accounts have zero followers. Our notebook instead adds +1 to the denominator (Laplace smoothing), which is mathematically cleaner and avoids the unvalidated assumption that zero-follower accounts are always fake. The cap approach could artificially inflate class separability.
- **following/posts ratio (Phase 3):** High values indicate an account that is active in following others but produces almost no content - a hallmark of spam bots. The same cap-at-max logic applies when posts = 0, with the same concerns noted above.

- **followers/posts ratio (Phase 4):** The paper notes that this feature has a lower correlation with the isFake label than the other two, yet the authors retain it, observing that it leads to better results - attributed mainly to MLP improvement. This finding highlights that features with low marginal correlation can still contribute to non-linear classifiers through interaction effects.

The paper's claim that the following/followers ratio is the most impactful single engineered feature is consistent with the cybersecurity literature on social-media bot detection, and is supported by the dramatic improvement observed in GNB and the partial recovery of SVM after its addition in Phase 2.

3.4 Redundancy Analysis

The three ratio features are calculated directly from the raw count features (numberOfFollowing, numberOfFollowers, numberOfPosts). This creates structural multicollinearity: numberOfFollowing appears in both the following/followers ratio and the following/posts ratio. The standard diagnostic for this in Logistic Regression would be the Variance Inflation Factor (VIF), though this was not computed in our reproduction. For a tree ensemble like Random Forest, correlated features are handled implicitly by random feature subsampling at each split - however, they can dilute feature importance scores, making interpretation less reliable.

The paper's correlation matrix shows that numberOfFollowers and followers/posts ratio have low correlation with the isFake label, while the following/followers ratio has the highest. To address redundancy formally, Recursive Feature Elimination (RFE) or PCA applied to the count-ratio block would identify the minimal sufficient feature set.

3.5 Meaningfulness of the Feature Engineering

The feature engineering is meaningful on both mathematical and cybersecurity grounds. Mathematically, ratio normalisation makes comparisons across accounts with vastly different activity levels fairer - a new account with 10 followers following 500 people is behaviourally more similar to a fake account than an established account with 10,000 followers following the same 500 people. From a threat-intelligence standpoint, the following/followers asymmetry is a well-documented indicator in the bot-detection literature. The digit-ratio in usernames captures the tendency of automated account-creation scripts to append random numeric strings (e.g., john_doe_48291).

3.6 Additional Features That Could Improve Performance

- **Account age / days since creation:** Newly created accounts with high following counts are disproportionately likely to be fake.
- **External URL and name == username:** Present in the Kaggle CSV but discarded during the merge due to the absence of equivalents in the InstaFake JSON. Prior work using these features achieved ~92.5% RF accuracy on the Kaggle dataset alone.
- **Posting frequency (posts / account age):** Bots often burst-post in short windows.
- **Follower growth velocity:** A sudden spike in followers is a strong indicator of purchased followers.
- **Biography text features:** Presence of promotional keywords, emoji density, language detection, or sentiment analysis of the biography text.

4. Reproducibility Analysis

4.1 Code Executability

Our notebook executes end-to-end without errors using standard Python libraries (pandas, numpy, scikit-learn, matplotlib, seaborn). All random operations use `random_state=42`, making results deterministic across runs.

4.2 Required Files and Dependencies

The four data files (`fakeAccountData.json`, `realAccountData.json`, `train.csv`, `test.csv`) are provided and correctly referenced. No external API calls or proprietary resources are required. The source paper does not publish its own code or exact preprocessing scripts, so our notebook constitutes an independent reimplementation based on the paper's description.

4.3 Hidden Preprocessing Steps

Several non-trivial preprocessing steps are implicit in the paper or diverge between the paper and our notebook:

- **Column name harmonization:** The InstaFake JSON uses verbose camelCase field names (e.g., `'userFollowerCount'`) while the Kaggle CSV uses terse notation (`'#followers'`). A `rename_map` handles this silently in the notebook.
- **Feature discarding:** The four Kaggle-only columns are dropped without explicit documentation in the notebook's markdown cells. A reader examining only the CSV headers would be surprised to find fewer features in the model.
- **nums/length username re-computation:** For the InstaFake JSON, this ratio is computed from raw digit count and username length. The Kaggle CSV already provides it pre-computed. The notebook handles both consistently, but the intermediate step is not highlighted.
- **Zero-denominator handling - cap vs. +1:** The paper caps ratio features at the maximum observed value when the denominator is zero. Our notebook adds +1 to all denominators. This produces different ratio values for accounts with zero followers or zero posts, which could explain part of the performance difference.
- **Duplicate removal.** 35 duplicate rows (1.85% of the dataset) were identified and removed prior to EDA, reducing the dataset from 1,890 to 1,855 records. The duplicates likely originated from overlap between the two source datasets. Removal had minimal impact on model performance but is important for preventing data leakage between train and test sets.
- **Incomplete model set:** The paper evaluates 7 classifiers across 4 phases (28 experiments). Our notebook evaluates all 7 classifiers (GNB, DT, LR, RF, MLP, KNN, SVM) on the full 10-feature set. The phase-by-phase ablation is not replicated.

4.4 Overall Reproducibility Assessment

The paper's core finding - that Random Forest outperforms other models on metadata-based fake account detection - is confirmed by our reproduction. However, the paper's specific numerical results (e.g., LR at 97% accuracy in Phase 4) cannot be reproduced because the exact dataset split, preprocessing code, and hyperparameter configuration are not published. The paper scores **medium reproducibility**: the methodology is described well enough to implement, but not in sufficient detail to obtain identical numbers. Publishing the data-processing code alongside the paper would significantly improve this.

5. Experimental Results

5.1 Experiments Performed

We replicated the paper's pipeline with three modifications:

1. All seven classifiers from the paper are trained (GNB, DT, LR, RF, MLP, KNN, SVM).
2. All three ratio features are added simultaneously rather than sequentially across phases.
3. The zero-denominator issue is handled by adding +1 instead of capping. The dataset is the same merged source (1,855 samples after deduplication), split 80/20 with stratification.

5.2 Evaluation Metrics

The following classification metrics are reported and interpreted in the context of fake account detection:

- **Accuracy** = $(TP+TN)/(TP+TN+FP+FN)$. Overall fraction of correct predictions. Intuitive but misleading under class imbalance - a classifier that labels everything as 'real' would achieve ~71% accuracy on our dataset.
- **Precision** = $TP/(TP+FP)$. Of accounts flagged as fake, the fraction that are actually fake. Low precision means innocent users are wrongly suspended - a reputational and legal risk.
- **Recall** = $TP/(TP+FN)$. Of all fake accounts, the fraction detected. Low recall means fake accounts operate undetected, causing ongoing harm. In a security context, recall is typically prioritised over precision.
- **F1 Score** = harmonic mean of precision and recall. Preferred over accuracy when classes are imbalanced. An F1 of 0.9 means that the model is both precise and sensitive.
- **AUC-ROC** - threshold-independent measure of discriminative ability. A value of 0.98 means that in 98% of randomly chosen (fake, real) pairs, the model assigns a higher probability to the fake account.
- **F_β Score (β=2)** weights Recall twice as much as Precision. In fake account detection, missing a fake account is more costly than wrongly flagging a real one, so Recall deserves higher weight.
- **Matthews Correlation Coefficient (MCC)** is a single balanced metric that accounts for all four cells of the confusion matrix. It is particularly reliable under class imbalance, ranging from -1 to +1, with 0 indicating random performance.

Metrics not computed in our notebook but present in the paper: **Log-loss** (measures probability calibration. The paper finds SVM log-loss ~10, indicating severe miscalibration) and per-phase metric deltas. These would be important to add in a complete reproduction.

5.3 Results

Model	Accuracy	Precision	Recall	F1	AUC-ROC
Gaussian Naive Bayes	0.4744	0.3586	1.0000	0.5278	0.9032
Decision Tree	0.9111	0.8220	0.8899	0.8546	0.9049
Logistic Regression	0.9164	0.8824	0.8257	0.8531	0.9661
Random Forest	0.9488	0.9018	0.9266	0.9140	0.9836
MLP	0.9326	0.8750	0.8991	0.8869	0.9825
KNN (k=5)	0.9137	0.8532	0.8532	0.8532	0.9594
SVM	0.9191	0.8762	0.8440	0.8598	0.9614

Random Forest outperforms the other models on all metrics, confirming the paper's primary claim. Its recall of 0.9266 means it detects ~93% of fake accounts in the test set (110 positive examples). The AUC-ROC of 0.9836 indicates near-perfect rank ordering. Logistic Regression achieves lower recall (0.8257) than Random Forest (0.9266), a gap corresponding to ~8 missed fake accounts in the test set. The paper reports LR reaching 97% accuracy in Phase 4 - our 91.5% is considerably lower, likely due to differences in dataset split and zero-denominator handling.

Comparison with the paper's reported results for related work: Our Random Forest F1 (0.9140) surpasses prior work using RF on the Kaggle CSV alone (~92.5% accuracy), and the best previously reported result (94% F1 with SMOTE oversampling) using RF. No SMOTE is applied in our reproduction, yet the result is competitive.

6. Conclusions

6.1 Key Findings

- Metadata-only fake account detection is highly effective: all three classifiers achieve >91% accuracy and >0.95 AUC-ROC.
- Random Forest is the best model, as the paper claims, achieving $F1 = 0.9140$ and $AUC-ROC = 0.9836$ in our reproduction.
- The paper's four-phase ablation framework is a valuable methodological contribution that our notebook does not fully replicate.
- The zero-denominator handling strategy (cap vs. +1) is a non-trivial design choice that affects results and should be standardised.

6.2 Lessons Learned

- Feature engineering encoding domain knowledge (bot behaviour) adds significant value even on a small, simple dataset.
- Publishing preprocessing code alongside the paper would substantially improve reproducibility, description alone is insufficient to obtain identical results.
- A single train/test split on a small dataset produces high-variance estimates, cross-validation should be the default.
- SVM's near-random performance ($AUC \sim 0.53$) highlights that not all classifiers are suitable for every dataset, kernel choice and class-weight adjustment are critical.

6.3 Strengths and Weaknesses of the Paper

Strengths: The four-phase ablation is methodologically rigorous and informative. The study covers seven diverse classifiers. The paper is transparent about dataset limitations and ethical concerns. The code-free description is nonetheless detailed enough to implement.

Weaknesses: No cross-validation, no hyperparameter tuning, no class imbalance handling, no code release, no statistical significance testing. The comparison to prior work is informal. Adversarial robustness is not discussed.

6.4 Suggestions for Future Improvements

- Apply five-fold stratified cross-validation to stabilise estimates.
- Tune hyperparameters (SVM kernel, RF depth, KNN k) via grid search.
- Apply SMOTE or class-weight adjustment to improve recall on the minority (fake) class.
- Publish the preprocessing and training code to enable exact reproduction.
- Conduct an adversarial robustness study: how much performance degrades when fake accounts strategically manipulate individual features.
- Add gradient-boosting models (XGBoost, LightGBM) and evaluate all seven paper classifiers for a complete reproduction.

7. Executive Summary

This project reproduces and critically evaluates the paper "Detection of Fake Instagram Accounts via Machine Learning Techniques" by Chelas, Routis, and Roussaki (Computers, MDPI, 2024). The paper proposes a supervised machine-learning pipeline for detecting fake Instagram accounts using publicly available account-level metadata.

The authors combine two datasets - the InstaFake JSON dataset and the Kaggle 'Instagram Fake Spammer Genuine Accounts' CSV - into a merged corpus of 1,890 samples (548 fake, 1,342 real). Prior to EDA, 35 duplicate rows were identified and removed, reducing the final dataset to 1,855 records. Eight base features are retained after harmonization, four Kaggle-only features are discarded. Three engineered ratio features (following/followers, following/posts, followers/posts) are added progressively across four experimental phases. Seven classifiers are evaluated at each phase. Random Forest is identified as the best performer.

Our reproduction confirms the paper's central finding: Random Forest achieves Accuracy = 94.9%, F1 = 91.4%, AUC-ROC = 0.9836 on our 80/20 stratified split. However, the paper's specific numerical results (e.g., Logistic Regression at 97% in Phase 4) cannot be reproduced, likely due to differences in dataset split and zero-denominator handling. Additionally, our notebook implements all seven paper classifiers. The phase-by-phase ablation is not replicated.

The paper's claim to outperform prior work using fewer features and smaller data is suggestive but not rigorously demonstrated - the comparison is informal, without controlling for dataset or evaluation protocol differences. The four-phase ablation is a genuine contribution. Key weaknesses include: absence of cross-validation, no hyperparameter tuning, no class imbalance handling, and no published code.

Error analysis (Random Forest, test set) reveals 8 false negatives - sophisticated fake accounts with profile pictures, longer biographies, and moderate follower counts - and 11 false positives - new legitimate users with a high following/followers ratio. These errors highlight the limits of metadata-only detection against well-crafted fake profiles.

8. Summing It Up

The Problem

Approximately 10% of Instagram's 2.4 billion accounts (240 million) are estimated to be fake, enabling fraud, phishing, spam, and influence operations at scale.

The Selected Source

Chelas, S., Routis, G., Roussaki, I. "Detection of Fake Instagram Accounts via Machine Learning Techniques." *Computers* 2024, 13(11), 296. DOI: 10.3390/computers13110296. Published 15 November 2024.

The Dataset

Two public datasets merged: InstaFake (JSON, 1,194 accounts) + Kaggle Instagram Fake Spammer (CSV, 696 accounts). Final: 1,855 samples (after removing 35 duplicates), 8 base features + 3 engineered ratio features.

The Methodology

Seven ML classifiers (GNB, RF, DT, LR, MLP, KNN, SVM) evaluated across four phases in which ratio features are added one at a time. Our reproduction trains all seven classifiers (GNB, DT, LR, RF, MLP, KNN, SVM) on the full feature set.

Main Findings of the Reproduction Study

- Random Forest is the best model - $F1 = 0.9140$, $AUC-ROC = 0.9836$. Confirmed.
- Paper's specific Phase 4 LR accuracy of 97% - not reproduced (our LR: 91.5%). Likely explained by different split and cap strategy.
- Phase-by-phase ablation not reproduced - features added all at once in notebook.
- All seven classifiers implemented. SVM achieves $AUC-ROC = 0.9614$ in our evaluation.

Were the Author's Claims Supported?

Partially. The core claim - that RF is best and metadata detection is effective - is confirmed. The claim of outperforming prior work is suggestive but not rigorously demonstrated and our reproduction does not fully validate the specific accuracy numbers.

Key Insights

- The following/followers ratio is the single most discriminative feature: fake accounts aggressively follow many users while attracting few followers.
- The four-phase ablation reveals that feature engineering, not dataset size, is the primary driver of performance improvements.
- False-negative fake accounts are 'high-effort' profiles - they have profile pictures, more posts, and longer biographies, mimicking real users.
- False-positive real accounts tend to be new users in an early audience-building phase, behaviourally indistinguishable from bots using only metadata.

Recommendation

This project is recommended as a solid starting point for metadata-based fake account detection. For production use, it should be combined with content analysis, graph-based network features, and temporal behavioural signals. The methodology would benefit from cross-validation, hyperparameter tuning, class-imbalance handling, and a standardised zero-denominator strategy before deployment.

Final Conclusion

Fake Instagram account detection using metadata features and classical machine learning is a well-supported, largely reproducible, and effective first-layer defence. The paper's central claims regarding Random Forest performance are confirmed by our experiments. The primary limitation - both of the paper and of our reproduction - is the gap between what metadata can capture and the sophistication of well-crafted fake accounts. A multi-layer detection system that combines the features

studied here with network topology and content-based signals would help catch more fake accounts while flagging fewer real ones.